

A top-down view of a grey desk with various items: a stethoscope, a laptop displaying a medical website, a tablet, a pair of glasses, a smartphone, a spiral notebook with a pen, and a coffee cup. A large white rounded rectangle is centered over the laptop, containing the course title and name. A faint watermark 'Licensed to Shreyans Salunke' is visible diagonally across the image.

Pharm*Achieve*

DIABETIC FOOT INFECTIONS

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

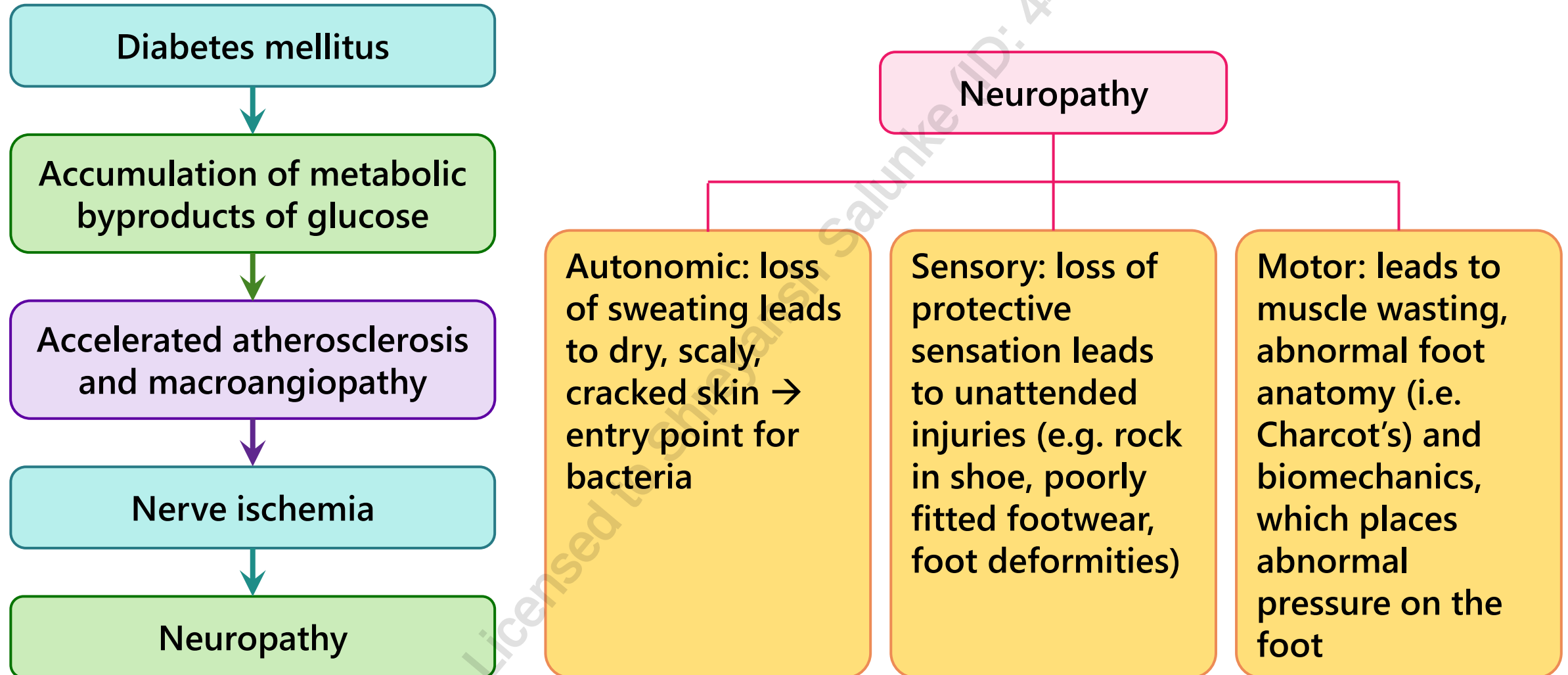
**Leadership &
Stewardship**

Professionalism

LEGEND

ADR	Adverse Drug Reaction
CLI	Critical Limb Ischemia
CrCl	Creatinine Clearance
CT	Computed Tomography
DFI	Diabetic Foot Infection
GI	Gastrointestinal
HbA1c	Hemoglobin A1c
MRI	Magnetic Resonance Imaging
MRSA	Methicillin-Resistant Staphylococcus Aureus
PAD	Peripheral Artery Disease
SIRS	Systemic Inflammatory Response Syndrome
SPICE	Serratia, Pseudomonas, Indole-Positive Proteus, Citrobacter, Enterobacter
TMP/SMX	Trimethoprim/Sulfamethoxazole
WBC	White Blood Cell Count

PATHOPHYSIOLOGY



PATHOPHYSIOLOGY

FYI

Vascular insufficiency

- Poor blood supply predisposes the feet to ulcers and impairs cellular-mediated wound healing

Immunopathy

- Impaired neutrophil function/phagocytosis

Physical deformities (e.g. Charcot foot or arthroplasty)

- Bones are weakened over time and changes in foot shape occur
- Joints collapse and foot takes on an abnormal shape such as flat bottom appearance (no plantar arch)
- Associated with neuropathy and increased disability

Patient behaviours

- Inadequate adherence to precautionary measures, foot inspection and hygiene procedures, excessive weight-bearing, as well as inappropriate footwear all increase infection risk

Patient disabilities

- Reduced patient vision and mobility may predispose those to diabetic foot infections

CLINICAL PRESENTATION OF INFECTION

FYI

Uninfected (PEDIS Grade 1)

No signs of infection
(e.g. purulence or inflammation)

Mild (PEDIS Grade 2)

≥ 2 symptoms of inflammation:
purulence, erythema,
pain, tenderness,
warmth, or induration

Cellulites/erythema ≤ 2
cm around the ulcer

Infection is limited to
the skin or superficial
subcutaneous tissue

Moderate (PEDIS Grade 3)

As above, plus ≥ 1 of:
cellulitis extending > 2
cm, lymphangitic
streaking, spread
beneath the superficial
fascia, deep-tissue
abscess, gangrene, and
involvement of muscle,
tendon, joint or bone,
fever

Severe (PEDIS Grade 4)

As above, plus systemic
toxicity/metabolic
instability: temp $> 38^{\circ}\text{C}$
or $< 36^{\circ}\text{C}$, chills,
tachycardia,
hypotension,
confusion, vomiting,
leukocytosis, acidosis,
severe hyperglycemia,
or azotemia

PEDIS CLASSIFICATION OF LESIONS

FYI

Perfusion	• Grade 1: no symptoms/signs of PAD • Grade 2: symptoms or signs of PAD, but not CLI • Grade 3: Critical limb ischemia (CLI)
Extent	• Size measured in cm²
Depth/tissue loss	• Grade 1: Superficial full-thickness ulcer • Grade 2: Ulcer penetrating to fascia, muscle, tendon • Grade 3: All subsequent layers of foot, including bone/joint
Infection	• Grade 1: no symptoms/signs • Grade 2: Inflammation of skin/sc only • Grade 3: Extensive erythema deeper than skin/sc • Grade 4: Systemic inflammatory response syndrome (SIRS)
Sensation	• Grade 1: No loss of protective sensation • Grade 2: Loss of protective sensation

RISK FACTORS

- Diabetics with poor glycemic control
- Vascular insufficiency
- Angiopathy
- Connective tissue disorders
- Neuropathy
- Minor trauma
- Ill-fitting shoes, nylons
- Charcot foot
- Previous ulcer/infection
- Obesity

DIAGNOSIS

- Semmes-Weinstein monofilament is used as a screening tool for neuropathy
- Blood glucose control/HbA1c
- Vascular assessment for peripheral pulse
- Orthopedic assessment for ulcers/deformities
- X-ray – assess for bone destruction, osteomyelitis, gas bubbles in soft tissue
- CT, MRI – assess for extent of disease, abscess, bone tissue involvement
- Bone scan, iridium scan – can differentiate between bone and soft tissue infection

GOALS OF THERAPY

- ✓ Treat current infection and wound
- ✓ Prevent occurrence and recurrent infections/ulcer
- ✓ Restore ability to use feet for daily living with appropriate shoes
- ✓ Maintain optimal glycemic control
- ✓ Prevent complications (e.g., osteomyelitis, gangrene, amputation)
 - ✓ About $\frac{1}{4}$ of skin ulceration will spread to deep tissue and/or bone
 - ✓ About 10-30% of diabetic foot ulcers progress to amputations

NON-PHARMACOLOGICAL MEASURES: PREVENTION

Proper glycemic control

Properly fitted/orthopedic footwear

- Snug fit is recommended, not too small or too large

Skin and nail care, avoid injury

- Check water temperature (e.g. with elbow) before putting feet in
- Check shoes before putting them on
- Do not walk barefoot, inspect feet daily for cuts/scratches/blisters
- Regular nail care with podiatrist/chiropractist
- Daily foot checks

Off-loading to avoid pressure: redistribution and relief of pressure

- Soft insoles, padding

Pressure relief with casts/crutches/braces

Foot baths are not recommended → can lead to maceration of skin

NON-PHARMACOLOGICAL MEASURES: TREATMENT

Proper wound care (use of appropriate dressings)

- Assess moisture of diabetic foot ulcer: exudates → use hydrocolloid dressing to absorb exudate (Aquacel®) versus low exudate → moist dressing (Allevyn®)
- Insufficient evidence to recommend a specific type of dressing (one that mimics physiological moisture and environment is recommended)

Sharp debridement to remove necrotic tissue and skin

Surgery

- Drainage of abscess, irrigation and debridement of infected wounds, amputation of gangrenous tissue
- Amputation proximal to foot is indicated for: rapidly progressing, life-threatening infection, chronic limb dysfunction/infection/deformity, other surgeries (e.g. such as bone reconstruction, arthroscopy or vascular reconstruction)

Maggot Debridement Therapy (available through special access program or clinical trials)

- Secretion of a proteolytic enzymes that dissolve tissue
- Regulated by Health Canada as drugs under the Food and Drugs Act and Regulations

COMMON PATHOGENS

Usually polymicrobial, especially in late stage, recurrent infection

- Common pathogens include aerobic gram-positive bacteria
 - *S. aureus, Streptococcus sp.*
- Gram-negative microbes are more prevalent in chronic or well-established infections
 - *Enterobacter, Klebsiella, Proteus, Pseudomonas, E. coli*
- Consider anaerobes if abscess, ischemic/devitalized/necrotic, foul-smelling tissue, ischemic wounds or deep tissue
 - *Bacteroides fragilis, Clostridium, Pepto-streptococcus*

DRUG THERAPY

- Specific treatment depends on prior therapy, severity and potentially culture if from deep/sterile site
 - Empiric therapy should always cover *S. aureus* and β -hemolytic streptococci
 - If necrotic tissue, foul-smelling discharge, or patient's feet exposed to excessive moisture, also cover gram-negatives and anaerobes
- Oral route for mild/superficial infections
- IV route if severe infection (deep space infection, osteomyelitis), no improvement from oral antibiotics within 48-72 hours, or systemic symptoms
- Consider osteomyelitis (radiographic imaging MRI>CT>X-ray) clinical signs like probe to bone
- Duration of antibiotics depend heavily on severity of infection and response to therapy
 - Mild infections: 7 days
 - Complicated/Severe infections: 14-21 days
 - Osteomyelitis: 4-6 weeks

DRUG THERAPY

Consider vancomycin, linezolid or daptomycin if MRSA suspected

Localized infection	• Usually cellulitis surrounding an ulcer, purulence at base of ulcer • Cover: aerobic gram-positives (group A/Beta-hemolytic <i>strep</i> and <i>S. aureus</i>) • Antibiotics: cloxacillin, amoxicillin-clavulanic acid, cephalexin >>> TMP/SMX, clindamycin, doxycycline
Extensive/complicated infection	• Extensive cellulitis, plantar abscesses, deep space infections • Cover: staphylococci, streptococci, anaerobes and common <i>Enterobacteriaceae</i> species • PO antibiotics: amoxicillin-clavulanic acid, moxifloxacin, TMP/SMX + metronidazole or clindamycin, ciprofloxacin/levofloxacin + metronidazole/clindamycin • IV antibiotics: cefoxitin, cephalosporins + metronidazole, 3rd gen cephalosporin + clindamycin, piperacillin/tazobactam, carbapenem (ertapenem if no <i>pseudomonas</i>)
Osteomyelitis	• IV antibiotics: piperacillin/tazobactam, carbapenem, 3rd generation cephalosporin + metronidazole or clindamycin • Long-term oral antibiotics can be considered e.g. TMP/SMX + metronidazole or clindamycin, ciprofloxacin/levofloxacin + metronidazole/clindamycin, among others

MONITORING PARAMETERS

EFFICACY ENDPOINTS		
PARAMETER	CHANGE	TIMEFRAME
Difficulty walking, pain	Improve Resolve	4-6 days 3 weeks
Ulcer	Improve Resolve	1-2 weeks 4-6 weeks
Purulent discharge, swelling, soreness	Improve Resolve	2-3 days 1-2 weeks
Elevated CrCl, WBC	Normalize	3-5 days
Fever	Normalize	24 hours
SAFETY ENDPOINTS		
ADR from antibiotic therapy e.g. nephrotoxicity, GI upset, allergic reaction	Prevent/minimize	Duration of therapy

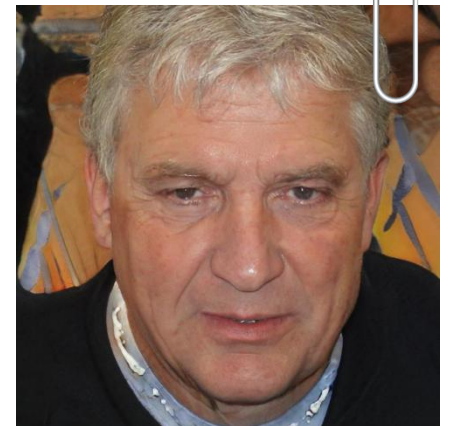
COUNSELLING TIPS ON HOW TO PREVENT DIABETIC FOOT INFECTION PREVENTIONS

- ☐ Wash and check your own feet daily
- ☐ Wear properly fitting shoes and socks
- ☐ Avoid open-toed shoes or shoes without socks
- ☐ Check shoes for foreign objects before putting them on
- ☐ Cut toenails correctly or have a professional involved
- ☐ See podiatrist for treatment of calluses/corns/thickened skin
- ☐ Get medical care for all bruises/cuts/scrapes/blisters
- ☐ Avoid heating pads and hot water soaks
- ☐ Wear slippers in places like public locker rooms/swimming pools to help avoid athlete's foot
- ☐ Achieve optimal blood sugar control
- ☐ Smoking cessation

CASE 1

AF is a 72-year-old male presenting to the emergency department with a history of several weeks of right foot pain/discharge. The day prior to coming to the hospital, he noted an increase in drainage and a few episodes of chills. On presentation, he has no fever or significant leukocytosis and is diagnosed with a grade 2 diabetic foot infection. There is no gas formation or foul smell from the wound. AF has a medical history of type II diabetes (HbA1c: 8.6%), hypertension, prostate cancer, Gleason grade 7, dyslipidemia, gout, visual impairment, r>l due to diabetic retinopathy, and diabetic neuropathy. He has no prior diabetic foot ulcers or infections. He currently takes anastrozole 1 mg PO daily, carvedilol 6.25 mg PO BID, allopurinol 200 mg PO daily, insulin glargine 22 units SC QHS, insulin lispro 6 units SC prior to meals, pravastatin 40 mg PO QHS, metformin 500 mg PO BID, and escitalopram 20 mg PO daily. He lives alone with several cats. He drinks socially and no longer smokes after quitting 10 years ago (previous 25 pack/year smoker).

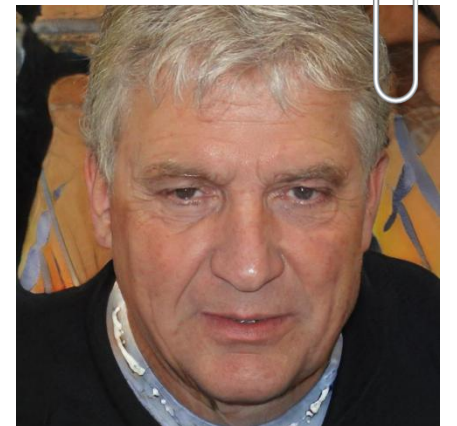
1. Which of the following components of diabetic foot infection management is most important for AF at this point?
 - a) Surgical revascularization
 - b) Pressure offloading
 - c) Wound care
 - d) Surgical debridement



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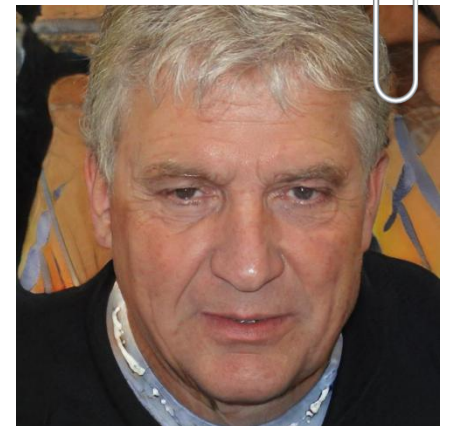
2. What empiric antimicrobial therapy would you suggest for AF?
- a) Vancomycin
 - b) Piperacillin/tazobactam
 - c) Ceftriaxone + metronidazole
 - d) Cefazolin



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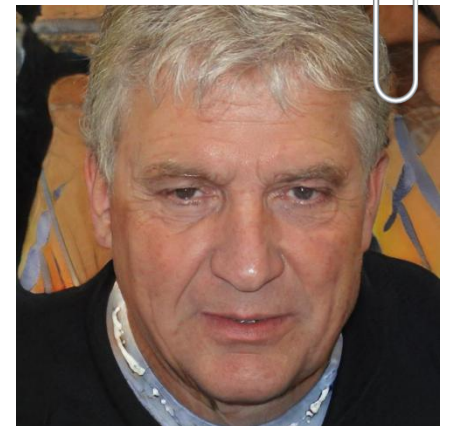
3. After two days of daily reassessment in the ER, AF remains stable. On review of his microbiology results, you notice his surgical debridement samples grew MRSA. What would you do?
- a) Change therapy to vancomycin
 - b) Change therapy to oral doxycycline
 - c) Add linezolid
 - d) Add tigecycline



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4. How would your therapy change if this were AF's third DFI in the past 6 months?
- a) Change therapy to levofloxacin + metronidazole
 - b) Change to piperacillin/tazobactam
 - c) Cover SPICE organisms using ertapenem
 - d) Add TMP/SMX suppressive therapy



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6. Diabetes Canada. Foot care: A step toward good health. <https://guidelines.diabetes.ca/GuideLines/media/Docs/Patient%20Resources/foot-care.pdf>.

CHANGE LOG

June 2025

- Slide 16: Added cefoxitin to IV antibiotics for extensive/complicated infections based on Diabetes Canada foot care guidance

October 2025

- Slides 5 and 6: Added 'FYI'
- Slide 10: Removed 'FYI'
- No content changes have been made

March 2026

- Slide 8: Removed slide on alternate diabetic wound classification as irrelevant for MCQ exam
- Slide 15: Minor update in osteomyelitis therapy